

Self-Learning: 80386 Memory Management & MESI Cache Protocol MPCA

Module 5

> **Definition:** The 80386 Memory Management Unit (MMU) provides two-stage address translation (**Segmentation & Paging**), while multiprocessor systems maintain cache consistency using the **MESI (Illinois) Protocol**.

1. 80386 Protected Mode Memory Translation

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1. Descriptor Tables (GDT, LDT, IDT):

- **GDT (Global Descriptor Table):** Holds segment descriptors accessible by all system tasks.

- **L**

2. Two-Level Paging Scheme (4 KB Pages):

A 32-bit Linear Address is split into:

...

2. MESI Cache Coherence Protocol

In symmetric multiprocessing (SMP) systems with private L1/L2 caches, every cache line exists in one of **four MESI states**:

...

M - M

State Transition Summary:

- **Read Hit:** Stays in M, E, or S state.

- **W**

3. Quick Recall Flow

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